

Phuoc Huynh

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WORK EXPERIENCE

Air Force Research Laboratory | AFRL Scholar

May 2026 – August 2026

- Created a Python link-margin analysis tool for hypersonic re-entry testbed telecommunications, raising model fidelity and enabling clean transfer across classification levels, adding **177s** of additional signal access time across original flight
- Simulated receiver link margin and re-entry trajectory in Ansys STK and MATLAB, validating telecommunications performance across every mission phase
- Optimized tool to produce most optimal latitude/longitude for any inputted receiver assets along any vehicle trajectory, increasing link margin dB by **85%**

Hawaii Corrosion Laboratory | Undergraduate Research Assistant

January 2026 – Present

- Designed on Onshape CAD and 3D printed a probe shield for cyclic corrosion tester, isolating the humidity/temperature sensor from corrosive cycling, decreasing corrodibility by **100%**
- Characterize surface and subsurface damage on SEM, 3D profilometer, ultrasonic sensor, and Raman microscope
- Acquire and analyze corrosion data across metal alloy samples for Dr. Lloyd Hihara's group, supporting graduate researchers and lab staff

State of Hawaii Department of Transportation | Engineer Intern

May 2025 – May 2026

- Co-led design of two cost-efficient security doors deterring after-hours loitering and cutting HVAC load, saving **\$80,000+ annually**
- Mapped 100+ HNL tenant lease areas in PROPWorks and 20+ inter-island terminal electrical panels in AutoCAD, citing meter installation points for accurate energy billing through Johnson Controls Metasys
- Consolidated 500+ tenant leases and permits across 15 airports into the engineering consultant's GIS database for HDOT/Oceanit tenant improvement initiative

PROJECT EXPERIENCE

Team Hokulele - Two-Stage Rocket | Ground Station & Avionics Engineer

September 2025 – May 2026

- Ran ground station operations and programmed flight computers through launch, holding telemetry lock across the entire flight
- Validated flight computer performance in vacuum chamber testing across simulated altitudes, confirming recovery deployment before launch
- Modeled PCB transmitter assemblies in SolidWorks, running material and weight trades to optimize avionics performance
- Competed at FAR Unlimited Launch Competition placing 8th out of 20 teams

ME 213 Engineering Design - Egg Retriever Robot | Mechanical Lead & Systems Engineer

August 2026 - Present

- Leading 5 subsystems (recovery, chassis, locomotion, circuits, software) as systems engineer, driving technical decisions and design cohesion across the full robot.
- Oversee management of team budget, schedule, resources, and overall project execution.

EDUCATION

University of Hawaii at Mānoa | BS in Mechanical Engineering | GPA: 3.42

Awards: University of Hawaii at Mānoa Academic Merit Scholarship

Coursework: Thermodynamics, Material Science, Engineering Design

ENGINEERING SKILLS

Analysis & Methods: Link Margin Analysis, Trajectory Simulation, Corrosion Analysis, Material Selection, Data Analysis

Software: Python, MATLAB, Ansys STK, SolidWorks, Onshape, AutoCAD, PROPWorks, Excel

Fabrication & Lab: 3D Printing, SEM, 3D Profilometry, Raman Microscopy, Ultrasonic Sensing, Vacuum Chamber Testing